

Ordane Kabengele Musau

ELECTRICAL & ELECTRONICS ENGINEERING | POWER SYSTEMS | EMBEDDED & IoT

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PROFESSIONAL SUMMARY

Final-year B.Eng. (Hons) Electrical and Electronics Engineering student with hands-on experience in industrial power systems, utilities, fault diagnosis, embedded systems, IoT and edge AI. Completed a 16-week technical internship at BRACONGO supporting power-factor correction, backup generation, motor troubleshooting and industrial cabling. CCNA and Lean Six Sigma Green Belt certified.

ENGINEERING EXPERIENCE

BRACONGO S.A. | Electrical Engineering Intern

Sep 2025 - Jan 2026

Direction Technique - Fluides & Energies

Kinshasa, DRC

- Supported assembly and installation of a 1,100 kVAr Schneider Electric power-factor compensation unit, including termination of 16 high-current 240 mm² copper conductors.
- Diagnosed a coolant-circulation airlock on a primary backup generator during an extended 14-day continuous run and helped implement a custom conduit solution to restore reliable cooling.
- Isolated motor single-phasing and intermittent contact faults in the filtration zone using voltage and continuity checks, supporting safe equipment restoration.
- Installed, labelled and verified 100+ m of power and shielded instrumentation cabling for brewing expansion, following electrical drawings, EMI-separation and cable bend-radius requirements.

Equity BCDC | Customer Service Intern

Sep 2022 - Nov 2022

Equity Leaders Program

Kinshasa, DRC

- Assisted customers with account registration and service requests while coordinating activities for the second Equity Leaders Program cohort.

SELECTED ENGINEERING PROJECTS

Smart Cold Chain Guardian | Group Design Project

2026

- Co-developed a last-mile cold-chain prototype integrating ESP32 sensing and actuator control with Raspberry Pi 5 edge computing; monitored temperature, humidity, CO₂/gas and motion and automated fan ventilation and servo louvres.
- Contributed to system integration and validation of a MobileNetV3-Small apple-quality classifier, achieving 99.7% Stage-1 and 88.3% four-class accuracy, with multimodal sensor fusion for condition assessment.

Acoustic Noise Energy Harvesting | Final Year Project

2026

- Designed and fabricated an electromagnetic acoustic harvester using a fixed-magnet/moving-coil mechanism and a 340+ turn SWG36 coil for comparison with piezoelectric transduction.
- Performed frequency-response, sound-pressure-level, air-gap, load-resistance and repeatability tests; characterised voltage, power and resonance behaviour and identified 2.5 Ω as the optimum tested load.

EDUCATION

Asia Pacific University of Technology & Innovation (APU)

Nov 2022 - Present

B.Eng. (Hons) Electrical and Electronics Engineering

Kuala Lumpur, Malaysia

TECHNICAL SKILLS

Software & Programming: MATLAB/Simulink, Python, C/C++, LTspice, Arduino IDE | Platforms: ESP32, Raspberry Pi, Arduino

Engineering: Power systems, industrial electrical systems, fault diagnosis, schematic interpretation, motors, sensors & instrumentation, experimental testing, data analysis, IoT and edge AI

CERTIFICATIONS & LANGUAGES

Certifications: Cisco Certified Network Associate (CCNA) | Lean Six Sigma Green Belt (CSSC)

Languages: French and Lingala (Native) | English (Fluent) | Swahili (Intermediate) | Tshiluba (Basic)